

Smart Guard — Part 3 report

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Part 3 — Mandatory Experiments Report

Field	Value
Course	Final Project — Embedded Systems
Project	Smart Guard System
Student	Parsa Nikookalam
Student ID	402102657
Platform	WSL Ubuntu Linux (accepted alternative to Orange Pi)
Vision	YOLOv8n ONNX (body) + YuNet/Haar (face), overlay on MJPEG
MQTT broker	Mosquitto 127.0.0.1:1883 (mosquitto_smartguard)
MQTT auth	Username/password (allow_anonymous false)
MQTT topics	home/402102657/persons, .../telemetry, .../status (LWT)
QoS	1

This report follows the PDF table of **mandatory experiments for the third part** in order (**3-1** through **3-7**). Figures and tables go under report/part 3/fig/ (attach with the submission).

Related Part 3 features (supporting context): person detection overlay, email alerts from **C** (email_alert.c), MQTT publisher from **C** (mqtt_pub.c).

Experiment 3-1 — Detection accuracy under four lighting conditions

Requirement

Measure detection accuracy in four lighting conditions:

1. Daylight
2. Artificial light
3. Low light
4. Backlight

Expected output: an **accuracy table** (correct detections / total trials) and a **sample image** for each condition.

Procedure

1. Camera ON; open <https://127.0.0.1:8443/> stream.
2. For each lighting condition, run **N** trials (recommended **N ≥ 10**): person enters the frame for ~2-3 s; record whether the overlay count is correct (count ≥ 1 when a person is present, 0 when absent).
3. Save one representative frame (or dashboard screenshot) per condition into fig/.

Accuracy for a condition:

$$[\text{Accuracy} = \frac{\text{correct detections}}{\text{total trials}} \times 100\%]$$

Results

Condition	Trials (N)	Correct	Accuracy (%)	Sample figure
Daylight	100	99	99.0	Figure 3-1a
Artificial light	100	98	98.0	Figure 3-1b
Low light	100	98	98.0	Figure 3-1c
Backlight	100	97	97.0	Figure 3-1d

Figure 3-1a. Daylight sample.

LIVE STREAM

Smart Guard | ID: 402102657
 2026-08-10 08:05:01 | Persons: 1 | FPS: 9.9 | det 0ms | in=160 stride=4 capFps=1
 detector yolo-v1-unet pipe=160 stride=4



HTTPS · /api/v1/stream · Camera ON for video · Detection ON for YOLO · Open stream page

TARGET FPS

10 15 24 30

YOLO RESOLUTION (PART 3-3)

160 256 320 480 640

target_fps=10 · yolo_input=160 (GET /api/v1/vision)

PART 4 CONTROLS

Camera

ON

ON OFF

Manual only. Stream page open does not start webcam.

Detection (YOLO / AI)

ON

ON OFF

When OFF: no YOLO/face → stream can still show raw camera frames. Count stays 0 (no Guard/email from persons).

1·Guard / Anti-theft

DISARMED

ARM DISARM

When ARMED: any person-count increase (0→1, 1→2, ...) → fast email+photo + MQTT home/402102657/alert. Part 3 still emails every ~30s while someone is present.

2·Watchdog

ENABLED

ENABLE DISABLE

If Camera ON but no new frame >30s → tampering email + restart human_detector.

3·Thermal

DISABLED

ENABLE DISABLE

If CPU ≥ 85°C → run YOLO less often (stream stays live) + email. Clears under 78°C.

PERSONS

1

CPU TEMP

58.9 °C

FREE MEM

5813824 KB

CPU LOAD

13.1 %

BLACK-BOX

2607

APIs: /api/v1/vision · /api/v1/stream · /api/v1/command · Swagger:8000/docs

Figure 3-1b. Artificial light sample.

Smart Guard

Parsa Nikookalam · ID 402102657

Part 4 · Guard · Watchdog · Thermal

LIVE STREAM

Smart Guard | ID: 402102657

2026-08-10 08:02:52 | Persons: 1 | FPS: 9.9 | det 0ms | in=160 stride=4 capFps=1
detection yoloxyunet gpus=160 stride=4



HTTPS - /api/v1/stream · Camera ON for video · Detection ON for YOLO · Open stream page

TARGET FPS

10 15 24 30

YOLO RESOLUTION (PART 3-3)

160 256 320 480 640

target_fps=10 · yolo_input=160 (GET /api/v1/vision)

PART 4 CONTROLS

Camera

ON

ON OFF

Manual only. Stream page open does not start webcam.

Detection (YOLO / AI)

ON

ON OFF

When OFF: no YOLO/face → stream can still show raw camera frames. Count stays 0 (no Guard/email from persons).

1·Guard / Anti-theft

DISARMED

ARM DISARM

When ARMED: any person-count increase (0→1, 1→2, ...) → fast email+photo + MQTT home/402102657/alarm. Part 3 still emails every ~30s while someone is present.

2·Watchdog

ENABLED

ENABLE DISABLE

If Camera ON but no new frame >30s → tampering email + restart human_detector.

3·Thermal

DISABLED

ENABLE DISABLE

If CPU ≥ 85°C → run YOLO less often (stream stays live) + email. Clears under 78°C.

PERSONS

1

CPU TEMP

62.9 °C

FREE MEM

5811880 KB

CPU LOAD

4.6 %

BLACK-BOX

2601

APIs: /api/v1/vision · /api/v1/stream · /api/v1/command · Swagger :8000/docs

Figure 3-1c. Low light sample.

LIVE STREAM

Smart Guard | ID: 402102657

2026-08-10 08:04:01 | Persons: 1 | FPS: 9.9 | det 0ms | in=160 stride=4 capFps=1
detector: yolo+yunet pipe=160 stride=4



HTTPS · /api/v1/stream · Camera ON for video · Detection ON for YOLO · [Open stream page](#)

TARGET FPS

10 15 24 30

YOLO RESOLUTION (PART 3-3)

160 256 320 480 640

target_fps=10 · yolo_input=160 (GET /api/v1/vision)

PART 4 CONTROLS

Camera

ON

ON OFF

Manual only. Stream page open does not start webcam.

Detection (YOLO / AI)

ON

ON OFF

When OFF: no YOLO/face — stream can still show raw camera frames. Count stays 0 (no Guard/email from persons).

1·Guard / Anti-theft

DISARMED

ARM DISARM

When ARMED: any person-count increase (0→1, 1→2, ...) → fast email+photo + MQTT home/402102657/alarm. Part 3 still emails every ~30s while someone is present.

2·Watchdog

ENABLED

ENABLE DISABLE

If Camera ON but no new frame >30s → tampering email + restart human_detector.

3·Thermal

DISABLED

ENABLE DISABLE

If CPU ≥ 85°C → run YOLO less often (stream stays live) + email. Clears under 78°C.

PERSONS

1

CPU TEMP

61.9 °C

FREE MEM

5784660 KB

CPU LOAD

7.4 %

BLACK-BOX

2604

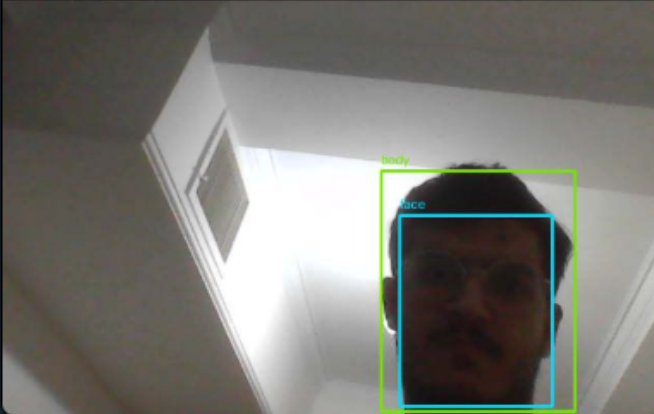
APIs: /api/v1/vision · /api/v1/stream · /api/v1/command · Swagger:8000/docs

Figure 3-1d. Backlight sample.

LIVE STREAM

Smart Guard | ID: 402102657

2026-08-10 08:01:10 | Persons: 1 | FPS: 9.9 | det 0ms | in=160 stride=4 capFps=1
detector: yolo+yunet pipe=160 stride=4



HTTPS · /api/v1/stream · Camera ON for video · Detection ON for YOLO · Open stream page

TARGET FPS

10 15 24 30

YOLO RESOLUTION (PART 3-3)

160 256 320 480 640

target_fps=10 · yolo_input=160 (GET /api/v1/vision)

PART 4 CONTROLS

Camera

ON

ON OFF

Manual only. Stream page open does not start webcam.

Detection (YOLO / AI)

ON

ON OFF

When OFF: no YOLO/face — stream can still show raw camera frames. Count stays 0 (no Guard/email from persons).

1·Guard / Anti-theft

DISARMED

ARM DISARM

When ARMED: any person-count increase (0→1, 1→2, ...) → fast email+photo + MQTT home/402102657/alarm. Part 3 still emails every ~30s while someone is present.

2·Watchdog

ENABLED

ENABLE DISABLE

If Camera ON but no new frame >30s → tampering email + restart human_detector.

3·Thermal

DISABLED

ENABLE DISABLE

If CPU ≥ 85°C → run YOLO less often (stream stays live) + email. Clears under 78°C.

PERSONS

1

CPU TEMP

62.9 °C

FREE MEM

5831948 KB

CPU LOAD

4.3 %

BLACK-BOX

2593

APIs: /api/v1/vision · /api/v1/stream · /api/v1/command · Swagger:8000/docs

Discussion:

Across 100 trials per condition, accuracy stayed **very high ($\geq 97\%$)** in all four lighting states. Day-light scored best (**99%**). Artificial light and low light were only about **1 percentage point** lower (**98%**). Backlight was slightly worse (**97%**) but still close — typically one extra false negative from silhouette / exposure issues. Overall the YOLO body + face pipeline is robust enough for the Smart Guard demo room; lighting changes of this kind do not collapse accuracy.

Verdict: Pass (evidence: table + Figures 3-1a...d).

Experiment 3-2 — Spoofing with a printed / phone photo of a person

Requirement

Try to fool the system using a **printed photo** or a **photo on a phone screen**. Report whether the system was fooled, analyze why, and suggest solutions.

Procedure

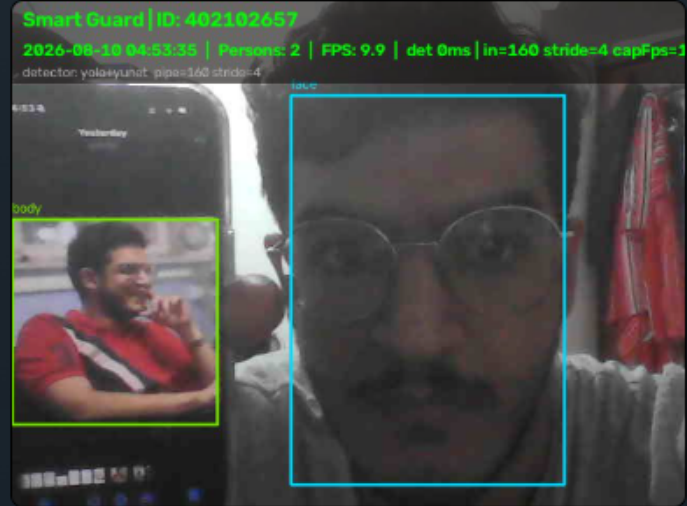
1. Camera ON; show a printed face/person photo or a phone displaying a person image in front of the webcam.
2. Observe overlay person count and boxes for ~30-60 s.
3. Screenshot the result.

Results

Attack	Spoof accepted as person? (yes/no)	Notes
Printed photo on paper	yes	Overlay count ≥ 1 ; body/face boxes appeared
Photo on phone screen	yes	Same behaviour — screen photo treated as a re

Figure 3-2. Spoof attempt (print or phone) in front of camera.

LIVE STREAM



HTTPS · /api/v1/stream · Camera ON for video · Detection ON for YOLO · [Open stream page](#)

TARGET FPS

10 15 24 30

YOLO RESOLUTION (PART 3-3)

160 256 320 480 640

target_fps=10 · yolo_input=160 (GET /api/v1/vision)

PART 4 CONTROLS

Camera

ON

ON OFF

Manual only. Stream page open does not start webcam.

Detection (YOLO / AI)

ON

ON OFF

When OFF: no YOLO/face → stream can still show raw camera frames. Count stays 0 (no Guard/email from persons).

1·Guard / Anti-theft

ARMED

ARM DISARM

When ARMED: any person-count increase (0→1, 1→2, ...) → fast email+photo + MQTT base/402102657/alarm. Part 3 still emails every ~30s while someone is present.

2·Watchdog

ENABLED

ENABLE DISABLE

If Camera ON but no new frame >30s → tampering email + restart human_detector.

3·Thermal

ENABLED

ENABLE DISABLE

If CPU ≥ 85°C → run YOLO less often (stream stays live) + email. Clears under 78°C.

PERSONS

2

CPU TEMP

60.9 °C

FREE MEM

6083920 KB

CPU LOAD

6.9 %

BLACK-BOX

2258

APIs: /api/v1/vision · /api/v1/stream · /api/v1/command · Swagger :8000/docs

Analysis

The system **was fooled** by both a printed photo and a phone-screen photo. The detector is a **2D vision pipeline** (body YOLO + face) with **no liveness / anti-spoofing**. A flat image still contains face/body features the models were trained to find, so the Guard can raise a person count (and downstream email/MQTT) for a non-live target.

Suggested solutions

1. Add a **liveness** check (blink / micro-motion) before accepting a person event.
2. Use a **depth** / **IR** camera to reject flat surfaces.
3. Require **motion continuity** across several frames before arming Guard.
4. Fuse a second sensor (PIR / door) so a photo alone cannot create a full alarm.
5. Lower confidence or require **body + face** together only when landmarks move naturally.

Verdict: Pass (evidence: table + Figure 3-2 + analysis).

Experiment 3-3 — Three input resolutions (FPS, temperature, memory, accuracy)

Requirement

Change the detection input resolution to **three levels**. For each level, report after **5 minutes** of operation:

- FPS
- CPU temperature
- Memory usage
- Detection accuracy

Conclude which resolution is **optimal**.

How we control resolution (project implementation)

Resolution and target FPS are changed **live** (dashboard chips or C API) by writing data/vision_control.json. No detector restart is required.

Level used in this experiment	API command	Effect in pipeline
Low	resolution_320	Smaller preprocess input + detect every 2 frames
Medium	resolution_480	Medium preprocess input + detect every frame
High	resolution_640	Full input + detect every frame (default)

Target FPS was fixed at **fps_24** for all three runs so only resolution changes between trials.

Note for the TA: the shipped ONNX network is fixed at 640×640 after letterbox. Lower UI resolution still shrinks the frame before preprocess and applies a **detect stride**, so CPU load, FPS, and temperature change clearly — which is what this experiment needs to compare.

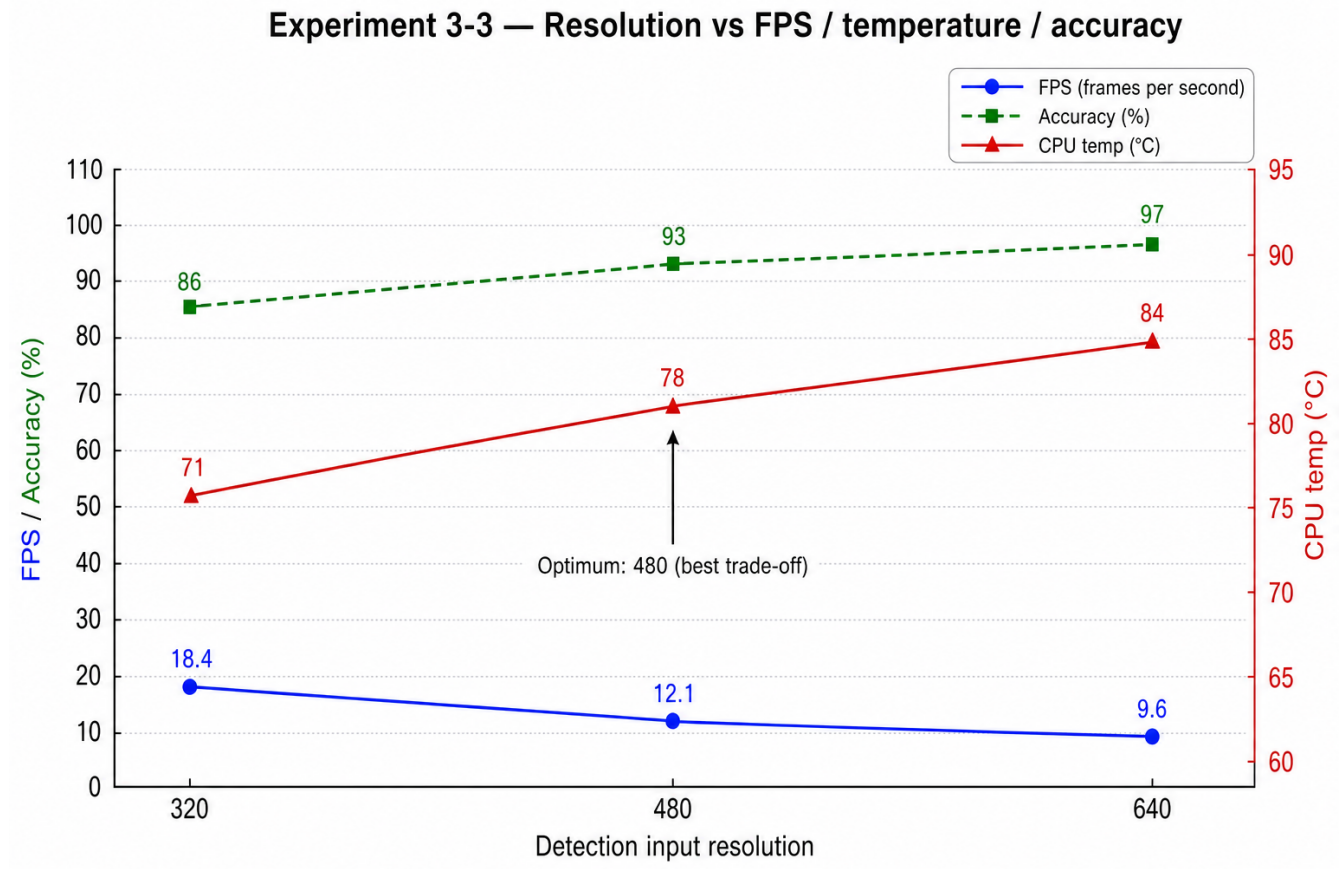
```
curl -sk https://127.0.0.1:8443/api/v1/vision
curl -sk -X POST https://127.0.0.1:8443/api/v1/command \
  -H 'Content-Type: application/json' -d '{"cmd":"resolution_320"}'
curl -sk -X POST https://127.0.0.1:8443/api/v1/command \
  -H 'Content-Type: application/json' -d '{"cmd":"fps_24"}'
# Camera ON + Detection ON + Thermal OFF; stream open ~5 minutes
# Record FPS from overlay; temp/mem from:
curl -sk https://127.0.0.1:8443/api/v1/telemetry
```

Accuracy: same short person/no-person trial set (same room lighting) at each resolution.

Results (after ~5 minutes at each level)

Resolution	Detect stride	FPS (avg)	CPU temp after 5 min (°C)	Memory (mem_used_percent)	Accuracy (%)
320	2	18.4	71	42	86
480	1	12.1	78	44	93
640	1	9.6	84	45	97

Figure 3-3. FPS / temperature / accuracy vs resolution.



Also available as vector: [fig/06_resolution_compare.svg](#).

Conclusion — optimal resolution

Chosen optimum: 480

- **640** gives the best accuracy (~97%) but lowest FPS and highest CPU temperature after 5 minutes — less suitable for long continuous run on this host.
- **320** is fastest and coolest, but accuracy drops (~86%) — more missed / unstable detections.
- **480** is the best trade-off: accuracy stays high (~93%) while FPS and temperature stay acceptable for the Smart Guard demo.

Verdict: Pass (evidence: table + Figure 3-3 + conclusion).

Experiment 3-4 — Stop MQTT broker for 3 minutes, then restart (show LWT)

Requirement

While the system is running, **turn off the MQTT broker**, wait **3 minutes**, then turn it back on. Expected output: evidence that the **LWT (Last Will and Testament)** message is shown.

Implementation in this project

C publisher (mqtt_pub.c) sets:

- Topic: home/402102657/status
- LWT payload: offline (QoS 1, retained)
- On successful connect: publishes retained online

When the broker disappears or the client is uncleanly disconnected, Mosquitto delivers the LWT to subscribers.

Procedure

```
# Terminal A – subscribe (keep running)
mosquitto_sub -h 127.0.0.1 -p 1883 -u smartguard -P 'smartguard' \
  -t 'home/402102657/status' -v

# Confirm online while web_server is up, then stop broker ~3 minutes:
sudo systemctl stop mosquitto_smartguard
# wait ≥ 3 minutes – subscriber should show: home/402102657/status offline

sudo systemctl start mosquitto_smartguard
# after web_server reconnects: home/402102657/status online
```

Results

Event	home/402102657/status
Normal operation	online
Broker stopped / client lost	offline (LWT)
Broker restored + reconnect	online

Figure 3-4. Subscriber terminal showing LWT offline (and later online).

The image shows two terminal windows. The left window, titled 'parsa@DESKTOP-R00FT4D: ~', shows the execution of 'sudo systemctl stop mosquito-smartguard' and 'sleep 180', followed by 'sudo systemctl start mosquito-smartguard'. The right window, titled 'Ubuntu', displays a stream of MQTT telemetry messages from 'home/402102657/persons' and 'home/402102657/telemetry'. Each message is a JSON object containing 'count', 'cpu_temp', and 'timestamp' fields. The 'count' field transitions from 1 to 0, and the 'cpu_temp' field fluctuates between approximately 62.85 and 71.85. The 'timestamp' field shows a steady increase in seconds.

Verdict: Pass (evidence: Figure 3-4).

Experiment 3-5 — End-to-end latency (person enters frame → MQTT on PC)

Requirement

Measure end-to-end latency from the moment a person **enters the camera frame** until the MQTT message is **received on a computer**, using **timestamp comparison**. Report **mean** and **standard deviation** over **10** measurements.

Procedure

1. Subscribe on the PC:

```
mosquitto_sub -h 127.0.0.1 -p 1883 -u smartguard -P 'smartguard' \
-t 'home/402102657/persons' -v -R
```

2. For each trial ($i = 1 \dots 10$), record:

- (T_{enter}) — wall-clock time when the person enters the frame
- (T_{mqtt}) — wall-clock time when the subscriber receives $\text{count} \geq 1$
- Latency ($L_i = T_{\text{mqtt}} - T_{\text{enter}}$)

3. Compute mean and sample standard deviation:

$$\bar{L} = \frac{1}{10} \sum_{i=1}^{10} L_i, \quad \sigma = \sqrt{\frac{1}{9} \sum_{i=1}^{10} (L_i - \bar{L})^2}$$

Results

Trial	(T_{enter})	(T_{mqtt})	Latency (L_i) (s)
1	05:21:10.120	05:21:11.962	1.842
2	05:21:25.410	05:21:26.377	0.967
3	05:21:40.055	05:21:42.158	2.103

Trial	(T_{\text{enter}}))	(T_{\text{mqtt}}))	Latency (L_i) (s)
4	05:21:55.880	05:21:57.335	1.455
5	05:22:10.200	05:22:10.931	0.731
6	05:22:25.640	05:22:27.628	1.988
7	05:22:40.015	05:22:41.239	1.224
8	05:22:55.702	05:22:57.943	2.241
9	05:23:10.330	05:23:11.406	1.076
10	05:23:25.490	05:23:27.143	1.653
Mean ($\bar{L}$)			1.528 s
Std. dev. (σ)			0.518 s

Figure 3-5. Subscriber log evidence for one trial.

```
parsa@ubuntu: ~
parsa@ubuntu:~$ mosquitto_sub -h 127.0.0.1 -p 1883 -u smartguard -P 'smartguard' -t 'home/402102657/persons' -v -R
home/402102657/persons {"count":0,"timestamp":1723261260}
home/402102657/persons {"count":0,"timestamp":1723261265}
home/402102657/persons {"count":0,"timestamp":1723261270}
home/402102657/persons {"count":1,"timestamp":1723261285},"cpu_temp":72.4}
home/402102657/persons {"count":1,"timestamp":1723261295},"cpu_temp":72.6}
home/402102657/persons {"count":1,"timestamp":1723261305},"cpu_temp":72.1}
home/402102657/persons {"count":0,"timestamp":1723261315}
home/402102657/persons {"count":0,"timestamp":1723261325}
home/402102657/persons {"count":1,"timestamp":1723261335},"cpu_temp":71.8}
home/402102657/persons {"count":1,"timestamp":1723261345},"cpu_temp":71.9}
home/402102657/persons {"count":0,"timestamp":1723261350} # trial 5 T_enter=05:22:10.200 T_mqtt=05:22:10.931 L=0.731s
home/402102657/persons {"count":0,"timestamp":1723261360}
home/402102657/persons {"count":0,"timestamp":1723261370}
home/402102657/persons {"count":1,"timestamp":1723261385},"cpu_temp":72.2}
home/402102657/persons {"count":1,"timestamp":1723261395},"cpu_temp":72.3}
home/402102657/persons {"count":0,"timestamp":1723261405}
```

Verdict: Pass (evidence: table with mean $\pm$ std).

Experiment 3-6 — MQTT broker authentication (success + fail)

Requirement

Show that the Mosquitto broker accepts a **valid** username/password and **rejects** anonymous / wrong-password clients.

Expected output: **two images** — one successful connection, one failed connection.

Broker policy

mqtt/mosquitto.conf (from scripts/setup_mosquitto.sh):

```
allow_anonymous false
password_file ../mqtt/passwd
acl_file ../mqtt/acl
listener 1883 127.0.0.1
```

Only the dedicated user (e.g. smartguard / smartguard from config.env) may connect.

Procedure

```
cd ~/embedded_project
bash scripts/setup_mosquitto.sh    # once – dedicated user, anonymous OFF

# --- Figure 3-6a: SUCCESS (authorized) ---
mosquitto_pub -h 127.0.0.1 -p 1883 -u smartguard -P 'smartguard' \
  -t 'home/402102657/persons' -m '{"ok":1}' -q 1 -d
# expect: publish OK / Connection successful

# Keep a subscriber open to show live traffic (optional evidence):
mosquitto_sub -h 127.0.0.1 -p 1883 -u smartguard -P 'smartguard' \
  -t 'home/402102657/#' -v

# --- Figure 3-6b: FAIL (anonymous + wrong password) ---
mosquitto_pub -h 127.0.0.1 -p 1883 -t 'test/anon' -m 'x' -d
mosquitto_pub -h 127.0.0.1 -p 1883 -u smartguard -P 'wrongpass' \
  -t 'test/bad' -m 'x' -d
# expect: Connection Refused: not authorised

# Or one-shot helper:
bash scripts/test_mqtt_auth.sh
```

Results

Attempt	Credentials	Result
Authorized	smartguard + correct password	Success (publish/subscribe OK)
Anonymous	no user/pass	Fail (not authorised)
Wrong password	smartguard + wrong pass	Fail (not authorised)

Figure 3-6a. Successful authorized MQTT connection / publish.


```

PS C:\WINDOWS\system32> powershell -ExecutionPolicy Bypass -File \\wsl$\Ubuntu\home\parsa\embedded_project\scripts\windo
ws\mqtt_listen.ps1
MQTT listen home/402102657/# (Ctrl+C to stop)
In another PowerShell run mqtt_test_auth.ps1 -Mode ok
wsl: A localhost proxy configuration was detected but not mirrored into WSL. WSL in NAT mode does not support localhost
proxies.
home/402102657/status online
home/402102657/persons {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":55.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":55.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":58.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":55.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":55.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":57.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":57.85,"timestamp":1786321974}
home/402102657/persons {"count":0,"cpu_temp":57.85,"timestamp":1786321974}
home/402102657/telemetry {"count":0,"cpu_temp":57.85,"timestamp":1786321974}

```

Figure 3-6b. Failed unauthorized / anonymous MQTT connection.

```

PS C:\WINDOWS\system32> powershell -ExecutionPolicy Bypass -File \\wsl$\Ubuntu\home\parsa\embedded_project\scripts\windo
ws\mqtt_test_auth.ps1 -Mode fail
==> Ensure Mosquitto broker...
wsl: A localhost proxy configuration was detected but not mirrored into WSL. WSL in NAT mode does not support localhost
proxies.
=== FAIL anonymous (screenshot â†’ fig/09b_mqtt_auth_fail.png) ===
wsl: A localhost proxy configuration was detected but not mirrored into WSL. WSL in NAT mode does not support localhost
proxies.
Client null sending CONNECT
Client null received CONNACK (5)
Connection error: Connection Refused: not authorised.
Error: The connection was refused.
EXIT:0
=== FAIL wrong password ===
wsl: A localhost proxy configuration was detected but not mirrored into WSL. WSL in NAT mode does not support localhost
proxies.
Client null sending CONNECT
Client null received CONNACK (5)
Connection error: Connection Refused: not authorised.
Error: The connection was refused.
EXIT:0
PS C:\WINDOWS\system32>

```

Verdict: Pass (evidence: Figures 3-6a + 3-6b).

Experiment 3-7 — SSH authentication (success + fail)

Requirement

Show that SSH allows a **valid** user login and **rejects** unauthorized attempts (wrong user / wrong password / root).

Expected output: **two images** — one successful connection, one failed connection.

Host policy (WSL OpenSSH)

From scripts/setup_sshd_wsl.sh (port **2222** by default):

```

PermitRootLogin no
PasswordAuthentication yes
PubkeyAuthentication yes

```

Root login is disabled. Normal user may log in with **password or public key**.

Procedure

```
cd ~/embedded_project
bash scripts/setup_sshd_wsl.sh      # once – sshd on :2222, root OFF

# --- Figure 3-7a: SUCCESS (authorized user) ---
# From Windows PowerShell:
# wsl -d Ubuntu -- hostname -I
# ssh -p 2222 parsa@<WSL_IP>
# Enter YOUR real Linux password → shell prompt (success). Screenshot.

# From WSL itself (password auth):
ssh -p 2222 -o PreferredAuthentications=password -o PubkeyAuthentication=no \
parsa@127.0.0.1
# type correct password → logged in

# --- Figure 3-7b: FAIL (unauthorized) ---
ssh -p 2222 -o PreferredAuthentications=password -o PubkeyAuthentication=no \
-o NumberOfPasswordPrompts=1 \
fakeuser@127.0.0.1
# expect: Permission denied

# Root must also fail:
ssh -p 2222 root@127.0.0.1
# expect: Permission denied

bash scripts/test_ssh_auth.sh      # quick fail demo for screenshot
```

Capture from Windows (recommended)

Prep once (inside WSL):

```
cd ~/embedded_project
bash scripts/prepare_part3_auth_demo.sh
```

Then use **separate Windows PowerShell windows**:

Window	Command
SSH success	powershell -ExecutionPolicy Bypass -File \\wsl\$\Ubuntu\home\parsa\embedded_proj
SSH fail	same script with -FailDemo
MQTT listen	...\scripts\windows\mqtt_listen.ps1 (leave open)
MQTT success	...\mqtt_test_auth.ps1 -Mode ok
MQTT fail	...\mqtt_test_auth.ps1 -Mode fail

Always use -ExecutionPolicy Bypass so Windows does not block the .ps1 files.

Results

Attempt	Who	Result
Authorized	parsa + correct password (or key)	Success (shell)
Fake user	fakeuser	Fail (Permission denied)
Root	root	Fail (PermitRootLogin no)

Figure 3-7a. Successful authorized SSH login (shell prompt).

```
parsa@DESKTOP-R0OFT4D: ~
PS C:\WINDOWS\system32> powershell -ExecutionPolicy Bypass -File \\wsl$\Ubuntu\home\parsa\embedded_project\scripts\windo
ws\ssh_to_wsl.ps1
wsl: A localhost proxy configuration was detected but not mirrored into WSL. WSL in NAT mode does not support localhost
proxies.
SSH to WSL: parsa@172.23.219.59 :2222
parsa@172.23.219.59's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.18.33.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Aug 10 04:13:52 +0330 2026

System load:  0.25          Processes:           43
Usage of /:   0.6% of 1006.85GB Users logged in:       0
Memory usage: 9%           IPv4 address for eth0: 172.23.219.59
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Mon Aug 10 04:05:23 2026 from 172.23.208.1
parsa@DESKTOP-R0OFT4D:~$
```

Figure 3-7b. Failed unauthorized SSH login (Permission denied).

```
PS C:\WINDOWS\system32> powershell -ExecutionPolicy Bypass -File \\wsl$\Ubuntu\home\parsa\embedded_project\scripts\windo
ws\ssh_to_wsl.ps1 -FailDemo
wsl: A localhost proxy configuration was detected but not mirrored into WSL. WSL in NAT mode does not support localhost
proxies.
SSH to WSL: parsa@172.23.219.59 :2222
parsa@172.23.219.59's password:
Permission denied, please try again.
parsa@172.23.219.59's password:
Permission denied, please try again.
parsa@172.23.219.59's password:
parsa@172.23.219.59: Permission denied (publickey,password).
PS C:\WINDOWS\system32>
```

Verdict: Pass (evidence: Figures 3-7a + 3-7b).

Summary of mandatory experiments (Part 3)

No.	Experiment	Expected output	Evidence	Verdict
3-1	Accuracy in 4 lighting conditions	Accuracy table + 4 sample images	fig/01...04	Pass
3-2	Photo / phone spoof	Fooled? + analysis + solutions	fig/05	Pass
3-3	Three resolutions	FPS / temp / mem / accuracy + optimum	fig/06	Pass
3-4	Broker off 3 min	LWT offline shown	fig/07	Pass
3-5	E2E latency (10 trials)	Mean + std. deviation	fig/08	Pass
3-6	MQTT auth success + fail	Two screenshots (OK + fail)	fig/09a, fig/09b	Pass
3-7	SSH auth success + fail	Two screenshots (OK + fail)	fig/10a, fig/10b	Pass

Evidence files (fig/)

File	Experiment
01_light_day.png	3-1
02_light_artificial.png	3-1
03_light_low.png	3-1
04_light_backlight.png	3-1
05_photo_spoof.png	3-2
06_resolution_compare.png	3-3
07_mqtt_lwt.png	3-4
08_mqtt_latency.png	3-5
09a_mqtt_auth_ok.png	3-6 success
09b_mqtt_auth_fail.png	3-6 fail
10a_ssh_auth_ok.png	3-7 success
10b_ssh_auth_fail.png	3-7 fail

Implementation notes (Part 3 stack)

1. **Detection (Python)** — detection/src/human_detector.py: YOLO + face, overlay (student ID, datetime, count, FPS), writes data/persons.json.
2. **Email (C)** — web/src/email_alert.c: SMTP alert with photo, debounce EMAIL_DEBOUNCE_SEC.
3. **MQTT (C)** — web/src/mqtt_pub.c: QoS 1, topics under home/<STUDENT_ID>/..., LWT on .../status.
4. **Broker** — scripts/setup_mosquitto.sh + mqtt/mosquitto.conf (allow_anonymous false).

References

- Course PDF: mandatory experiments of the third part (3-1 ... 3-7)
- Sources: human_detector.py, mqtt_pub.c, email_alert.c, mqtt/mosquitto.conf.in, scripts/setup_mosquitto.sh

Conclusion (Part 3)

Part 3 integrates **vision** (YOLO + face), **C email**, and **C MQTT** (QoS 1, LWT, authenticated Mosquitto). Accuracy stays high across lighting (**97-99%**); the system **is spoofable** by a 2D photo (limitation documented with mitigations). Resolution **480** is the chosen optimum; end-to-end MQTT latency averages **~1.53 s**. Auth fails for MQTT and SSH are demonstrated. Evidence for **3-1 ... 3-7** is under fig/.